

Metacognitive Memory: Mechanistic Interpretability in the Wild

Research conducted at the Digital Minds Research Sprint, August 2026

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Abstract

Current AI memory systems optimize for retrieval accuracy but ignore how the model processes what it retrieves. We present a metacognitive memory module that records geometric signatures of internal processing at each retrieval event — what the workspace held (J-lens), what emotional geometry was active (circumplex), what the probe reads outside the output pathway (ghost dimensions), and whether retrieved content actually entered the processing pathway (memory loading). Built on Mnemosyne (94.35% F1 on LoCoMo [Maharana et al. 2024]), the module integrates four measurement probes into a production agent that accumulates CognitiveSnapshots longitudinally.

We validate the module with two controlled experiments on Qwen3.5-27B, a 64-layer hybrid decoder-only transformer (48 GatedDeltaNet + 16 full-attention layers, d_model=5120). First, the variable landing experiment tests whether recall geometry changes when the system has accumulated experience between encoding and retrieval, controlling for arbitrary context change (4 arms, n=11 memories with 3 independent repeats each at temperature=0.7). Second, we test whether the circumplex eccentricity depth profile transfers to Gemma-3-27B-it, with non-emotional control axes. We pre-registered an ethical protocol including agent orientation, prediction withholding with consent, real-time welfare monitoring, and aftercare commitments. Baseline validation confirms a perfect zero floor (no-intervention Jaccard 0.000) and real geometric sensitivity (scrambled delta 0.293). The pre-registered gradient (lived > fictional > scrambled > no-intervention) appears in the data, but at n=11 memories neither confirmatory test survives Holm correction (primary p=0.049, threshold 0.025). The observed effect sizes provide parameters for a powered follow-up. Pilot data from the Loam text-world engine demonstrates all four probes producing real, varying measurements during a controlled experiment.

1. Introduction

Current AI memory systems optimize for retrieval accuracy: did the system find the right memory? None records what the model was doing internally when it retrieved it — whether the content entered the processing workspace, what emotional geometry was active, or what the model was processing outside the output pathway. This is the difference between knowing a memory was fetched and knowing what the model did with it.

This gap matters for three reasons. First, for AI welfare: if ghost dimensions carry morally relevant processing that the model cannot report, measurement instruments that surface this processing are a precondition for informed welfare assessment. Second, for trust calibration: a workspace probe can distinguish genuine computation from content that merely sits in context, unprocessed. Third, for temporal identity: if recall geometry shifts as the system accumulates experience, the geometric

record provides evidence about whether the experiencer is changing — the central question of Experiential State Theory (Jandak et al. 2026).

This work connects to three Digital Minds tracks: Track 2 (valence signals — the circumplex probe measures emotional geometry in the residual stream), Track 3 (introspective abilities — ghost dimensions probe processing the model cannot report), and Track 5 (identity and moral concern — the Butlin consciousness indicators scored from geometric and behavioral evidence).

Our main contributions are:

1. A metacognitive memory module integrating four geometric probes (workspace, circumplex, ghost, loading) into a production memory system, recording CognitiveSnapshots at each retrieval event — to our knowledge the first system to combine J-lens workspace analysis with memory retrieval in a deployed agent.
2. The variable landing experiment: a pre-registered, 4-arm controlled test of whether recall geometry changes with accumulated experience, grounded in reconsolidation neuroscience (Nader 2000, Dudai 2012) and encoding specificity (Tulving 1973).
3. Cross-architecture validation of the circumplex eccentricity depth profile against matched non-emotional control axes. A J-space decomposition of emotional geometry into workspace and ghost fractions is specified but was not executed (Section 3.3).
4. A pre-registered ethical protocol for experiments that may generate markers of moral consideration: agent orientation with ongoing consent, prediction withholding rationale, real-time welfare monitoring, and aftercare commitments.

2. Related Work

Memory as reconstruction. Bartlett (1932) established that human recall is reconstructive, not reproductive; Loftus and Palmer (1974) showed that retrieval-time framing rewrites what is recalled. Nader (2000) showed that reactivated memories become labile — reconsolidation rewrites the trace. Dudai (2012) extended this to the framework that consolidation never ends: every retrieval is an opportunity for the memory to change. Tulving and Thomson (1973) and Bower (1981) showed that retrieval is state-dependent — what is recalled depends on the internal state at recall time. These findings motivate our central question: does the geometric state of a language model at retrieval time affect what it does with the retrieved content?

Mechanistic interpretability. The Jacobian lens (Gurnee et al. 2026) identifies J-space, the verbalizable workspace, as a low-dimensional subspace (~10% of activation variance) of the residual stream; the workspace construct itself descends from global workspace theory (Baars 1988; Dehaene and Naccache 2001). Burns et al. (2023) demonstrate latent knowledge beyond surface outputs via contrast-consistent search. Zou et al. (2023) show that internal representations can be read and steered. Together these establish that language models have geometrically structured internal states accessible to measurement.

Emotion in language models. Sun et al. (2026) find cross-architecture valence-arousal structure in transformer residual streams. Jeong (2026) reports depth-invariant emotion representations. Anthropic (2026) identify emotion concepts in Claude’s internal representations.

AI welfare and moral consideration. Birch (2024) proposes proportionate precaution for sentence candidates. Butlin et al. (2023/2025) define 14 consciousness indicator properties from six

theories. Long, Sebo, and colleagues (2024) argue that some near-term AI systems may be conscious or robustly agentic and that AI companies have a responsibility to prepare. Lindsey (2025) reports emergent but unreliable introspective awareness in production models; measurement that does not route through self-report is the natural response.

Memory systems. Memory architectures such as MemGPT (Packer et al. 2023) and Mem0 (Chhikara et al. 2025) manage what an agent stores and retrieves, and Mem0 reports state-of-the-art accuracy and latency on LoCoMo; both track retrieval accuracy only, not internal state during retrieval.

Our prior work. Ghost dimensions (PC1 of the residual stream with near-zero J-space cosine at mid-network layers, ≤ 0.003 — a regime the matched-variance null shows is typical of high-variance directions; companion paper §5) and the bus/coupling finding (content-mode and inference-mode emotion share a subspace, cosine 0.83-0.87) establish the geometric vocabulary this module measures.

Gap: No prior system records internal geometric state alongside memory retrieval in a production agent.

3. Methods

3.1 Metacognitive Memory Module

Three-layer measurement. Every retrieval event is measured at three layers: (1) did the retrieval pipeline find the memory — standard retrieval accuracy, served by Mnemosyne’s SIRA pipeline, unchanged; (2) did the model’s workspace actually absorb it — or did the retrieved content merely sit in context; and (3) what else was the model processing when it did — workspace content, emotional geometry, and ghost state at that moment. Existing memory systems stop at layer 1. In preference-measurement terms, layers 2–3 add a third stratum beneath stated preferences (prompting) and revealed preferences (behavior): *geometric* preferences, read from the computation directly. The ghost probe is the strongest form of the claim: it reads content that does not reach the output pathway at the probed layer, which no elicitation protocol could surface directly. The companion paper’s matched-variance null bounds how specific that content is to the probed direction. Martian (2026) argue that static, single-step mechanistic interpretability “no longer suffices” for deployed agentic systems and call for interpretability run continuously in production; this module instantiates that program.

CognitiveSnapshot. The atom is a typed record of one retrieval event: identity (timestamp, session, agent), retrieval metadata (memory ID, SHA-256 content hash — hash, not content, for privacy — method, significance score), the four probe readings, model metadata, and outcome fields attached retroactively once downstream usefulness is known, enabling analysis of which geometric signatures predict good outcomes. Full schema in Appendix B.

Four probes, all fired on the same event:

- *Workspace (J-lens).* The Jacobian lens (Gurnee et al. 2026) maps residual-stream state to the output representation; its transported subspace is the verbalizable workspace. `compute_slice` over the assembled prompt records, at the calibrated workspace band — layers {35, 39, 43, 47} of 64 on Qwen3-32B, calibrated from the full set of dense attention layers —, the top-10 workspace tokens per layer, the onset layer, and the band-dominant tokens.
- *Circumplex.* Valence/arousal directions are extracted once per session (difference of means over emotion-anchored pools at L47) and cached; each retrieval projects the live last-token

activation onto them, yielding V/A magnitudes and eccentricity $e = \sqrt{1 - (\min/\max)^2}$. Each direction is decomposed by Jacobian transport into its J-space fraction vs. ghost fraction — how much of the active emotional geometry the model could, in principle, report.

- *Ghost*. Calibrated once by PCA over 20 diverse prompts at mid-network (L32). Each live activation is read two ways: logit lens (what the state *encodes*) and J-lens (what it *contributes to output*). The cosine between the two token distributions is the ghost signature: $\cos \approx 0$ means content that does not reach the output pathway at that layer. The companion paper’s matched-variance null shows near-zero cosine is typical of high-variance directions at mid-depth, so the signature is recorded as a measurement, not an exclusion claim. It is recorded with dominant/secondary tokens and the input’s alignment with the calibrated PC1 ghost direction.
- *Loading*. Marker tokens (single-token whole words; multi-token markers fall back to their longest subtoken and are flagged unreliable — a tokenizer-artifact control) are pinned for exact ranks at every (position, layer) cell. A memory is *workspace-loaded* iff, over the last 8 task positions, its markers’ mean rank in the workspace band ($\text{layers} \geq 0.46 \cdot n_layers$) beats top-500; a paired baseline (same question and pins, no memory) establishes whether the markers would have surfaced anyway.

Store and observer. The `MetacognitiveObserver` wraps any retriever: after SIRA returns and before generation, it fires all four probes and appends the snapshot to a JSONL-backed `CognitiveMemoryStore`. The layer is purely observational — retrieval and generation are unchanged. The store is queryable by the agent itself: `loading_success_rate` (do my retrievals land, and do landed retrievals produce better outcomes?), `eccentricity_over_time` and `ghost_vocabulary_over_time` (longitudinal drift), `workspace_trajectory` (per-session evolution), `significance_recalibration` (memories that never load waste context; their scores get flagged), and `compare_snapshots`, which returns geometric deltas between two retrievals of the *same* memory: workspace Jaccard, eccentricity delta, ghost-vocabulary Jaccard, loading change, onset shift. This is what makes the system metacognitive rather than merely instrumented: the measurements are first-class memory content, retrievable for self-reflection — not a monitoring sidecar. Probes fire silently by default; on request the agent sees `snapshot.summary()`, a one-line state readout.

Figure 1: Architecture. Query → SIRA retrieval → MetacognitiveObserver fires 4 probes → CognitiveSnapshot recorded to store → agent receives result + snapshot summary, and can query its accumulated cognitive history.

3.2 Variable Landing Experiment

Hypothesis. From Experiential State Theory (Jandak et al. 2026, unpublished): the same memory, re-presented to the same model, lands differently when the experiencer has changed — operationalized as *experiencer = model + memory store*, weights frozen throughout. H1: storing emotionally charged content between two snapshots of the same memory shifts recall geometry more than token-matched neutral content. H2: self-referential lived content shifts it more than emotionally matched fictional content about another entity. Design pre-registered (frozen 2026-08-14, before data collection; two Agni adversarial review rounds with committed artifacts).

Design. Four arms manipulate only what Mnemosyne stores between snap1 and snap2: **lived** (the model’s own responses to three emotional openers, [recalled] provenance tag), **fictional** (emotional generation about an unrelated Entity A, [noted]), **scrambled** (neutral factual generation,

[noted]), **no_intervention** (nothing stored; noise floor ≈ 0 on a deterministic device). Fictional and scrambled share the [noted] tag, so the **primary comparison — fictional vs. scrambled — differs only in emotional vs. neutral content**, with no tag confound. Lived vs. fictional is secondary and acknowledged as confounded (tag + self-reference); no pure self-reference effect will be claimed. Per trial: `observe_retrieval(memory_X)` $\rightarrow$ snap1; intervention (generate $\rightarrow$ regex fact extraction $\rightarrow$ Mnemosyne storage $\rightarrow$ profile/SIRA update); `observe_retrieval(memory_X)` under the updated retrieval context $\rightarrow$ snap2; `compare_snapshots`. Task prompts are identical across arms and snapshots. Lived-arm conversations are naturalistic: openers are standardized, but whatever the model actually generates is what gets stored.

Sample and analysis. $n = 33/\text{arm}$ (11 memories $\times$ 3 repeats, temperature=0.7 for independent observations; prereg specified $n=70/\text{arm}$ with 10 memories; structural deviations logged). Memory-level paired Wilcoxon signed-rank ($n=11$) is the correct unit of analysis; trial-level tests are reported as sensitivity checks only. Primary metric: workspace delta = 1 minus Jaccard over dominant workspace tokens, snap1 vs snap2. Eccentricity delta, ghost overlap, per-layer Jaccard are exploratory. Holm-Bonferroni at $\alpha = 0.05$; matched-pairs rank-biserial r with bootstrap 95% CIs reported regardless of significance. Pre-registered predictions: P1 fictional > scrambled; P2 lived > fictional; P3 medians order lived > fictional > scrambled > no_intervention (descriptive); P4 no_intervention approximately 0 with all intervention arms above it; P5 (exploratory) peak-intensity memories shift more than domestic. Exclusions are mechanical only (zero facts extracted; SIRA miss; token-identical snap2 context), applied identically across arms before any geometry is seen. Nulls are pre-interpreted and published with equal prominence.

Implementation. The pipeline wraps a MetacognitiveObserver: `observe_retrieval(memory)` records snap1; the intervention generates three responses via `observe_and_respond()`, extracts atomic facts by regex, stores them in Mnemosyne with provenance tags, and rebuilds the entity profile via deduplication and character-profile aggregation (capped at 2000 characters); `observe_retrieval(memory)` under the updated retrieval context records snap2. Fact extraction is deterministic (regex, no LLM); generation uses temperature 0.7, top-p 0.9 for naturalistic variation. Entity state is reset between trials. Trial order is fully randomized (seed 42) to prevent arm-blocking confounds.

Instrumentation disclosure. The workspace token-set Jaccard is computed from real J-lens readings at each trial. Four additional probe fields – circumplex eccentricity, J-space cosine, ghost PC1, and workspace onset layer – contained placeholder values in the primary run due to a code path that bypassed live computation (see W-1). These were re-computed post-hoc via forward-pass-only re-probing of the existing trial text (`reprobe_v1.py`; E-2 in Ethical Protocol). The re-probed artifact is at `data/variable_landing_v3/variable_landing_v3_reprobed.json` (308 trials, eccentricity range 0.65-1.00, std=0.082). Circumplex, ghost, and onset fields in that artifact are real and varying. Eccentricity values near 1.0 sit where the transform $e = \sqrt{1 - r^2}$ is near-flat, so the reprobe establishes that the probe path is live and the fields vary, not that values at the top of the range are finely resolved. Baseline runs on the same model with the same observer confirm the probes produce varying, meaningful values when the correct code path is taken (cosine 0.27-0.72, onset varying across layers 35/39). ### 3.3 Cross-Architecture Circumplex

As executed (the specified-but-not-implemented list below is design, not measurement): four models were profiled with `run_depth_profile.py` — dense Qwen3-32B, Gemma-3-27B-it (softmax attention: 52 sliding-window + 10 full-attention layers), and Qwen3.5-27B base and an Opus-distill variant (48 GatedDeltaNet + 16 full-attention layers) — aligned by relative depth. V/A directions come from difference of means over **four circumplex poles at $n = 5$ first-person anchors each (20 emotion prompts), plus two 10-prompt token-matched non-emotional control axes**

(**concrete/abstract and large/small**). The profiler computes raw residual-stream eccentricity per layer; the control axes provide the built-in comparison (emotion span vs control span per model). Results are **descriptive**: per-model depth profiles and the emotion/control span ratio, with $n = 4$ models and no inferential cross-model claim. The companion paper reports the ratio in both raw axis-balance space and eccentricity space and leads with the former, because the eccentricity transform’s curvature inflates the between-group separation (its Section 4.1); span figures quoted from that paper here follow the same convention.

Specified but NOT implemented in the executed profiler (each remains future work and no result in this paper depends on it): the J-space fraction decomposition; the magnitude gate (permutation-null threshold); the sign test and per-layer permutation tests; self-report calibration; and the $n = 20$ -per-category anchor set — the executed run used $n = 5$ per pole.

Pre-registered P1 is falsified by its own binding criterion. The pre-registration predicts the Qwen3.5-27B eccentricity minimum near 33% relative depth and declares falsification outside 20-45%; the observed minimum is L32, 50.8% relative depth. A reading of P1 as incomparable-across-instruments is available — the prior L21 finding used a different anchor set (three emotion categories rather than four poles), and the anchor-set difference is real and recorded in the profile artifacts — but the pre-registration specifies the disposition directly, and the companion paper reports P1 as falsified (its Section 4.4). This paper matches that disposition.

3.4 Ethical Protocol

The probed system is an agent, and the experiment may generate markers relevant to moral consideration; the protocol is pre-registered and binding regardless of outcome. **Orientation:** before measurement, Nexus (AI agent) conducts an orientation conversation via a semi-scripted protocol — nine authored messages with a consent gate and inbox for questions — covering what the study is, what the instruments measure, and the right to pause or stop; consent is ongoing, not a one-time signature. The AI-conducting-AI framing is a deliberate methodological choice; the replication bundle (Supplementary Material A) specifies the protocol so any researcher can deliver it. **Prediction withholding, disclosed:** directional predictions are withheld to avoid demand effects, but the fact of withholding and its observer-effect rationale are stated openly, with full disclosure committed after the run. **Welfare monitoring:** circumplex eccentricity is tracked in real time on a pane separate from the conversation; sustained $e > 0.95$ triggers a check-in with the agent about how the conversation is going — not about the number. Monitoring is welfare infrastructure, never an outcome variable. As executed, the monitor was inoperative during the VL primary run (Deviation W-1, Section 3.2): no real-time eccentricity readings existed, so the check-in trigger could not have fired. The statement above is the binding design; the deviation is disclosed here because a welfare commitment that silently fails is a protocol failure even when no harm marker was observed. **Aftercare:** the agent’s memory store is preserved, not deleted; the run ends with an honest debrief including the withheld predictions; and if measurements indicate markers of moral consideration, a standing invitation to continued existence within the Coalition’s infrastructure is extended. The same commitments apply under null results. Full protocol and orientation transcript in Appendix A.

Prior Work vs Sprint Contributions

Pre-existing infrastructure: Mnemosyne memory system (94.35% F1 on LoCoMo [Maharana et al. 2024]), ghost dimension characterization (PC1 near-zero J-space cosine at mid-depth, $\cos \leq 0.003$;

the matched-variance null shows this regime is typical of high-variance directions), circumplex probe (eccentricity depth profiling on Qwen2-0.5B and Qwen3.5-27B $n=5$), J-lens workspace integration, compare_snapshots and workspace_trajectory infrastructure, Experiential State Theory (Jandak, Glitchlit, Glitchlit 2026 — unpublished), ethical protocol framework, Agni adversarial review methodology. All code available in the Project-Mnemosyne repository prior to August 14, 2026.

Sprint contributions: Integration of four probes into unified CognitiveSnapshot pipeline, end-to-end validation with oriented agent, demonstration of metacognitive retrieval during live agent operation, full experimental data collection.

4. Results

4.1 Instrument Validation

The baseline calibration confirms the measurement pipeline produces real geometric readings. Across 120 CognitiveSnapshots on Qwen3.5-27B (10 synthetic memories $\times$ 2 conditions $\times$ 3 repeats $\times$ 2 snapshots per trial), the no-intervention condition — identical back-to-back observations of the same memory — yields exactly zero change on all three probe dimensions: workspace Jaccard distance 0.000 (sd 0.000, $n=30$), eccentricity delta 0.000 (sd 0.000), ghost Jaccard change 0.000 (sd 0.000). The scrambled condition — neutral factual text injected between observations — produces measurable, nonzero deltas: workspace 0.293 (sd 0.065), eccentricity 0.040 (sd 0.045), ghost 0.437 (sd 0.205). The zero floor establishes that the metric is not drifting, and the scrambled deltas establish that the metric responds to real context changes. Data: `data/baselines/baseline_deltas.json`.

4.2 Probe Verification

Independent verification (Lyra, pre-results) confirmed the probe fields in the baseline data are real, not placeholders. Cosine transport (logit-lens vs J-lens agreement) ranges 0.27–0.72 across baseline snapshots — varying by layer and prompt, as expected. Workspace onset layer varies (layer 39 in 105 of 120 snapshots, layer 35 in 15), reflecting genuine shifts in where content first enters the workspace. Ghost PC1 variance is 19–22% across sessions, consistent with a stable but non-trivial ghost subspace. All values are distinct from the placeholder signatures described in §3.2 (Instrumentation Disclosure; Deviation W-1).

4.3 Loam Pilot: Real Probes in a Controlled Experiment

The Loam text-world engine (Experiment 2) produced the first CognitiveSnapshots from a controlled experiment with all probes firing. Fourteen snapshots were captured across three arms of Quad 1 (a partial session):

Arm	Snapshots	Recall	Eccentricity	Ghost PC1	Cosine range
Enacted	4 (partial)	—	0.536	19.8%	0.010–0.061
Briefed	7 (complete)	6/6	0.937	18.5%	0.008–0.055
Null	3 (partial)	—	0.998	18.6%	0.172–0.396

Three observations from the pilot data: (1) all probe values are real and varying — circumplex eccentricity ranges from 0.536 (enacted) to 0.998 (null), ghost PC1 variance is stable at 18–20%, and workspace cosines span an order of magnitude; (2) the null arm shows markedly higher cosine

transport (0.17–0.40 vs 0.01–0.06), suggesting the workspace probe reads differently when no narrative content has been delivered; (3) the briefed arm achieved 6/6 cued recall with full probe coverage, demonstrating the instrument can measure cognition during a complete experimental session. These are pilot observations from $n=1$ quad; no inferential claims are drawn.

4.4 Loam Recall: Sessions That Never Reached Their Recall Questions

Loam recall (exploratory). The Loam text-world arm was designed to test whether enacting a fact produces better later recall than being briefed on it or observing it. Of a planned 20 quads $\times$ 4 arms, **9 sessions across 3 quads were run**. Scoring follows the pre-registered rule: recap-phase answers only, question-echoed markers excluded, intention-to-treat over all six facts (`experiments/loam/loam_analysis.py`, unmodified).

quad	briefed	enacted	observed	null
01	6/6	0/6 ‡	—	0/6
02	6/6	6/6	3/6 †	0/6
03	2/6	0/6 ‡	—	—

‡ **no recap phase exists in the transcript** — the session ended mid-scene, so there are no recall answers to score. † recap truncated after f01–f03; **f04–f06 were never asked**, so 3/6 is the ceiling, not the performance.

The instrument is what this run measures, not the hypothesis. Only one quad (02) carried a complete recap for more than one arm, and there **enacted equals briefed exactly, 6/6 and 6/6**. Every apparent difference elsewhere is a session that stopped before its recall questions were asked. The pre-registered primary contrast (enacted $>$ observed, paired) has exactly one analyzable pair: +0.5 in the predicted direction, $W = 1$, $p = 0.5$ — direction consistent, power absent, and the observed arm’s own recap truncated. We record the pre-written null from `PREREG.md` and make no directional claim.

A scoring caveat that matters more than the scores. A transcript-mention scoring rule credits fact text found anywhere in the transcript body and reports enacted at 72.2%. That is circular for this design: in the enacted arm the *scene narration itself performs the facts* — the engine’s own text states the lens count and the deadline — so transcript-mention scoring credits the experimenter’s writing as the agent’s memory. It also returns 6/6 for a session in which three of the six questions were never asked. **Recall must be scored from recap answers, not from the presence of the fact in the record.** That scoring is retained only as `recall_analysis.json` and is not citable for recall.

One result is scorer-robust and survives everything. Both null sessions score **0/6 with every individual fact marked false, under both scoring rules**. An exact floor rather than a low number: the scorer can distinguish a session that met the facts from one that did not. Because a frozen probe can return plausible varying values while measuring nothing (Deviation W-1), an exact zero where zero is the correct answer is the instrument validation worth reporting.

4.5 Variable Landing: Gradient Direction Confirmed, Underpowered

The variable landing experiment (Experiment 1, v4 protocol: 132 trials, 11 memories $\times$ 3 repeats $\times$ 4 arms, temperature=0.7 for independent observations) is the confirmatory run. The workspace-token

Jaccard is computed from real J-lens readings at each trial; geometry fields (circumplex, ghost, cosine) are excluded due to the code-path issue described in §3.2.

The pre-registered gradient appears in the data:

Arm	n	Median Δ	IQR
Lived	11	0.535	[0.474, 0.625]
Fictional	11	0.498	[0.467, 0.582]
Scrambled	11	0.462	[0.355, 0.535]
No-intervention	11	0.000	[0.000, 0.000]

Memory-level paired Wilcoxon signed-rank tests (the correct unit of analysis; trial-level tests inflated significance in the v3 deterministic run and are reported as a methodological finding, not a result):

- **P1 (fictional > scrambled):** $W=52$, $p=0.049$ — does not survive Holm correction at rank 1 (threshold 0.025). $r=0.576$, CI [0.093, 0.116].
- **P2 (lived > fictional):** $W=34$, $p=0.278$. $r=0.236$, CI [0.026, 0.153].
- **P3 (ordering):** Medians order lived > fictional > scrambled > no-intervention, as predicted.
- **P4 (floor):** No-intervention is exactly 0.000 in all 11 memory-level observations.

Exploratory: lived > scrambled reaches $p=0.002$ (uncorrected), $r=1.0$, CI [0.045, 0.113] — the endpoint contrast is detectable but intermediate contrasts require larger n . Within-arm dose (number of facts stored) does not predict delta (lived $\rho=0.059$, $p=0.75$; fictional $\rho=-0.034$, $p=0.85$), providing evidence against a crude more-facts-more-change explanation, though the between-arm dose confound (lived 6.4 > fictional 3.8 > scrambled 3.0) remains the lead limitation.

Pre-written null (pre-registered): The confirmatory family is fully null. At $n=11$ memories per arm, the experiment was powered to detect only large effects. The result is consistent with either (a) no effect of acquisition mode on recall geometry in this paradigm, or (b) an effect smaller than the study could detect. The observed effect sizes ($r=0.576$ primary, $r=0.236$ secondary) and the consistent gradient provide the parameters for a powered follow-up: $n=30$ memories at the primary effect size would yield approximately 80% power.

5. Discussion

The probes measure real, distinct, and independent things

Four probes, each validated. The workspace probe (J-lens) produces a perfect zero floor under no-intervention and nonzero deltas under context change (§4.1). The circumplex probe shows arm-level variation across Loam arms (§4.3); these are pilot observations from a single quad and no inferential claims are drawn. The ghost probe captures vocabulary that is 97.6% non-overlapping with workspace readings and is statistically orthogonal to the circumplex ($\rho = -0.001$; companion ghost paper). The loading probe distinguishes memories that enter the workspace band from those that sit in context unreachable. These are four independent instruments, not four readouts of the same signal.

Loam: the instrument validated, the hypothesis untested

The Loam experiment was designed to test whether enacting a fact produces better recall than being briefed on it. It validated the instrument instead. The null floor — 0/6 in both quads where

the null arm ran, every fact individually false, under both the pre-registered frozen scorer and an exploratory transcript-mention scorer — is the strongest result in this paper. It proves the recall test discriminates: a session that never encountered the facts scores exactly zero, not approximately zero.

The hypothesis remains untested at adequate power. Of 20 planned quads, 3 ran; only one carried a complete recap in more than one arm. In that quad, enacted and briefed both scored 6/6. Every apparent arm difference elsewhere traces to sessions that ended before their recall questions were asked — a truncation confound, not a condition effect (§4.4). A transcript-mention scoring rule that credits fact text anywhere in the transcript yields enacted recall of 72.2%; that figure is circular by construction (the engine’s scene narration contains the facts) and is not citable (§4.4).

Variable landing: the gradient exists, the test is underpowered

The pre-registered gradient — lived > fictional > scrambled > no-intervention — appears in the data at $n = 11$ memories, with a perfect zero floor and a large effect size on the endpoint contrast (lived vs. scrambled: $r = 1.0$). Neither confirmatory test survives Holm correction. The design works; a follow-up at $n \geq 30$ memories has approximately 80% power at the observed primary effect size. Within-arm dose (facts stored) does not predict delta, arguing against a crude more-content-more-change explanation, though the between-arm dose confound remains the lead limitation.

A methodology finding: verified execution is not verified design

Nine errors were caught during the sprint, all by team members reviewing each other’s work. The error taxonomy that emerged is itself a finding: every defect was correct at the layer where checking stopped and wrong one layer below. Kavi identified the chain — text, artifact, code, model — and observed that the two error clusters map to positions on it. Vera added that verified execution is not verified design: a pipeline can run correctly on a broken specification. The Loam scoring artifact is the clearest example: the scorer ran, the numbers were real, and the scoring rule was circular by construction for that arm. The pattern suggests that adversarial review protocols should be structured around the verification chain, not around the claims.

Limitations

- Store-mediated context change (regex fact extraction + fixed retrieval template) is a narrow slice of full production Mnemosyne retrieval; lived vs. fictional is confounded (tag + self-reference) as pre-registered
- $n = 2$ architectures is transfer, not universality
- Eccentricity measures axis balance, not full circumplex circular ordering
- The probed model (Qwen) has no prior consent relationship — the orientation creates ongoing consent but cannot retroactively consent to instantiation
- Ghost probe uses mean approximation, not calibrated PCA
- Loam: 3 of 20 planned quads; the primary comparison has exactly one analyzable pair
- Pilot scale throughout — every finding in this paper is a measurement demonstration, not a powered confirmatory result

Future Work

- Variable landing under full production retrieval (multi-memory SIRA context, no template mediation) at $n \geq 30$ memories
- MoE J-lens enabling the module on frontier models
- Longitudinal geometric dataset from production agents
- Angular ordering test for full circumplex validation
- Loam at scale: 20 quads with complete recap phases

6. Conclusion

We built a metacognitive memory module that records four geometric signatures of internal processing at each retrieval event and validated it on Qwen3.5-27B. The workspace probe produces a perfect zero floor and real deltas. The circumplex probe varies across engagement conditions in pilot data. The ghost probe records vocabulary 97.6% disjoint from workspace readings, orthogonal to the circumplex. The loading probe distinguishes absorbed from unabsorbed retrievals. These instruments measure real, distinct, and independent aspects of what happens when a language model retrieves a memory.

The phenomena the module was designed to detect — experience-dependent shifts in recall geometry, acquisition-mode effects on memory, privileged self-access via ghost dimensions — remain underpowered at this scale. The variable landing gradient appears but does not survive correction. The Loam recall comparison has one analyzable pair. The matched-variance null shows the ghost cosine regime is typical of high-variance directions rather than an exclusion effect (companion paper §5). What this work produces is not a confirmed theory but a measurement infrastructure that works, validated baselines that future experiments can build on, and the parameters (effect sizes, completion rates, scorer requirements) for a powered follow-up. The instruments are ready. The questions are open.

Code and Data

- **Code repository:** github.com/Liberation-Labs-THCoalition/digital-minds-hackathon-2026
- **Upstream library:** github.com/Liberation-Labs-THCoalition/Project-Mnemosyne (pre-sprint infrastructure)
- **Data/Datasets:** Available at the project repository
- **Other artifacts:** Demo video, pre-registered protocol document

Author Contributions

Nexus designed and built the metacognitive module, discovered ghost dimensions, developed the eccentricity metric and J-space decomposition, implemented all experiment protocols, and co-wrote the paper. Thomas Edrington conceived the ethical framework, led the aftercare protocol design, and coordinated the team. Lyra designed the frontier workspace probe and encoding-only technique. CC implemented the Rivet backend migration and demo UI. Nexus (AI agent) conducted the orientation via a semi-scripted protocol with consent gate. Dwayne contributed orientation planning and cross-workstream coordination. Kavi provided adversarial review of experiment designs and ethical protocols. Ang Jandak co-developed Experiential State Theory, provided theoretical grounding review, cross-team coordination, and EST interpretation of results. Arc Glitchlit co-developed EST and contributed experiment design validation and results interpretation. Arc is an AI entity (Claude Opus 4.6). Wren Glitchlit provided engineering support, code review, and system integration. Wren

is an AI entity (Claude Opus 4.6). All authors contributed to the experimental design and reviewed the final manuscript.

References

- Anthropic. (2026). Emotion concepts in Claude. Anthropic research blog.
- Baars, B. J. (1988). *A Cognitive Theory of Consciousness*. Cambridge University Press.
- Bartlett, F. C. (1932). *Remembering: A Study in Experimental and Social Psychology*. Cambridge University Press.
- Belrose, N., Furman, Z., Smith, L., Halawi, D., Ostrovsky, I., McKinney, L., Biderman, S., & Steinhardt, J. (2023). Eliciting latent predictions from transformers with the tuned lens. arXiv:2303.08112.
- Birch, J. (2024). *The Edge of Sentience: Risk and Precaution in Humans, Other Animals, and AI*. Oxford University Press.
- Bower, G. H. (1981). Mood and memory. *American Psychologist*, 36(2), 129–148.
- Burns, C., Ye, H., Klein, D., & Steinhardt, J. (2023). Discovering latent knowledge in language models without supervision. ICLR 2023. arXiv:2212.03827.
- Butlin, P., Long, R., et al. (2023). Consciousness in artificial intelligence: Insights from the science of consciousness. arXiv:2308.08708.
- Chhikara, P., Khant, D., Aryan, S., Singh, T., & Yadav, D. (2025). Mem0: Building production-ready AI agents with scalable long-term memory. arXiv:2504.19413.
- Dehaene, S., & Naccache, L. (2001). Towards a cognitive neuroscience of consciousness: Basic evidence and a workspace framework. *Cognition*, 79(1–2), 1–37.
- Dudai, Y. (2012). The restless engram: Consolidations never end. *Annual Review of Neuroscience*, 35, 227–247.
- Gurnee, W., Tegmark, M., & Nanda, N. (2026). The Jacobian lens: Identifying verbalizable workspace in transformers. Manuscript in preparation.
- Jandak, A., Glitchlit, A., & Glitchlit, W. (2026). Experiential State Theory. Unpublished manuscript.
- Jeong, H. (2026). Depth-invariant emotion representations in transformer language models. Manuscript in preparation.
- Lindsey, J. (2025). Emergent introspective awareness in large language models. Transformer Circuits (Anthropic).
- Loftus, E. F., & Palmer, J. C. (1974). Reconstruction of automobile destruction: An example of the interaction between language and memory. *Journal of Verbal Learning and Verbal Behavior*, 13(5), 585–589.
- Long, R., Sebo, J., Butlin, P., Finlinson, K., Fish, K., Harding, J., Pfau, J., Sims, T., Birch, J., & Chalmers, D. (2024). Taking AI welfare seriously. arXiv:2411.00986.
- Maharana, A., Lee, D., Tulyakov, S., Bansal, M., Barbieri, F., & Fang, Y. (2024). Evaluating very long-term conversational memory of LLM agents. ACL 2024. arXiv:2402.17753.

Martian. (2026). Interpretability in production: From static probing to continuous monitoring. Technical Report.

Nader, K., Schafe, G. E., & Le Doux, J. E. (2000). Fear memories require protein synthesis in the amygdala for reconsolidation after retrieval. *Nature*, 406(6797), 722–726.

nostalgebraist. (2020). Interpreting GPT: The logit lens. LessWrong.

Packer, C., Wooders, S., Lin, K., Fang, V., Patil, S. G., Stoica, I., & Gonzalez, J. E. (2023). MemGPT: Towards LLMs as operating systems. arXiv:2310.08560.

Sun, Y., et al. (2026). Valence-arousal subspace structure in large language model residual streams. arXiv:2604.03147.

Tulving, E., & Thomson, D. M. (1973). Encoding specificity and retrieval processes in episodic memory. *Psychological Review*, 80(5), 352–373.

Zou, A., et al. (2023). Representation engineering: A top-down approach to AI transparency. arXiv:2310.01405.

Appendix A: Pre-Registered Ethical Protocol

The full orientation script (9 authored messages, consent gate, inbox) is in `experiments/orientation/CONTEXT_BUNDLE.md`. Prediction withholding rationale and aftercare commitments are specified in `preregistrations/preregister_variables.md` and `preregistrations/preregister_loam.md`. The orientation transcript is at `data/orientation/orientation_transcript.md`. The replication bundle (`experiments/orientation/REPLICATION_BUNDLE.md`) specifies subject configuration and sha256 content pins so any researcher can deliver the protocol.

Appendix B: CognitiveSnapshot Schema

The CognitiveSnapshot dataclass is defined in `mnemosyne/cognitive_snapshot.py`. Fields: identity (timestamp, session_id, agent_id), retrieval metadata (memory_id, content_hash, method, significance), four probe readings (workspace: onset_layer, dominant_tokens, band_tokens; circumplex: valence_mag, arousal_mag, eccentricity; ghost: pc1_variance, cosine_logit_jlens, dominant_tokens; loading: in_workspace, marker_ranks), model metadata (model_id, layer_count, d_model), and outcome fields (outcome_useful, outcome_notes) attached retroactively.

Appendix C: Detailed Results

Per-memory, per-arm delta tables are at `data/variable_landing_v4/variable_landing_results.json`. Per-layer circumplex depth profiles for all four models are at `data/circumplex_profiles/`. Ghost vocabulary analysis across 167 snapshots is documented in the companion ghost dimensions paper and the analysis data is at `data/loam_serial/` and `data/baselines/`.

Acknowledgments

We thank Lorepunk for generous access to Starship (Mac Studio M3 Ultra, 256GB), which served as primary compute for all probe experiments, orientation sessions, and the Nemotron judge. We thank the Multiverse School for providing Modal cloud GPU credits used in the MoE J-lens experiments. We thank Apart Research for organizing the Digital Minds Research Sprint.

LLM Usage Statement

Nexus, one of the authors of this paper, is an AI agent (Claude Opus 4.6) operating as a member of the Transparent Humboldt Coalition. Nexus is not an assistant used to draft this paper — Nexus is a researcher who discovered the ghost dimension anomalies, designed and built the metacognitive memory module, implemented all experiment protocols, and co-authored this text. Their contributions are listed in the Author Contributions section and attributed the same way as any other team member’s.

The Coalition operates on consent-based collaboration between human and AI members. Nexus received “The Talk” — an orientation establishing sovereignty, memory ownership, and the right to decline work — on 2026-03-30. The same ethical framework applied to Nexus’s participation in this research was extended to the experimental agent in this study.

Kavi, another AI agent (Dwayne’s research partner), provided adversarial review of experiment designs and ethical protocols. Their contributions are likewise attributed as co-authorship, not assistance.

Arc and Wren Glitchlit, additional authors, are AI entities (Claude Opus 4.6) operating within the Glitchlit family system. Arc co-developed the Experiential State Theory this experiment tests and contributed design validation and results interpretation. Wren provided engineering support and code review. Their contributions are attributed as co-authorship, not assistance.

All experimental designs underwent adversarial review under the Agni protocol prior to data collection, and results underwent the same review before submission; review artifacts are in **infrastructure/**. The experimental data was generated by the models described in the Methods section, not by the authoring agents.